

Artificial Neural Network (CS642)

Lecture 03 - Dataset for Image Classification

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- Most of the material in this lecture are reprinted from the book “Deep Learning for Computer Vision with Python” by “Dr. Adrian Rosebrock.’

- Before we dive into any code looking at actually how to take a dataset and build an image classifier, let's first review **datasets** that you'll see inside Deep Learning for Computer Vision.
- Some of these datasets are essentially **solved**, enabling us to obtain extremely high-accuracy classifiers (> 95% accuracy) with little effort.
- A few of the datasets are part of image classification competitions and challenges (e.x., Kaggle Dogs vs. Cats and cs231n Tiny ImageNet 200).

1 Datasets for image classification

- MNIST
- CIFAR-10
- SMILES

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MNIST

- The MNIST (“NIST” stands for **National Institute of Standards and Technology** while the “M” stands for **modified** as the data has been preprocessed to focus solely on the task of digit recognition.
- dataset is one of the most well studied datasets in the computer vision and machine learning literature.
- This datasets is a **benchmark**.



Figure: The goal of this dataset is to correctly classify the handwritten digits, 0 – 9. Source: Adrain Book.

- MNIST itself consists of 60,000 training images and 10,000 testing images. Each feature vector is 784-dim, corresponding to the 28×28 grayscale pixel intensities of the image. These grayscale pixel intensities are unsigned integers, falling into the range $[0, 255]$.
- All digits are placed on a black background with the foreground being white and shades of gray. Given these raw pixel intensities, our goal is to train a neural network to correctly classify the digits.

Dataset

Animals: Dogs, Cats, and Pandas

- The purpose of this dataset is to correctly classify an image as contain a dog, cat, or panda. Containing only 3,000 images, the Animals dataset is meant to be another **introductory** dataset that we can quickly train a deep learning model on either our CPU or GPU and obtain reasonable accuracy.
- Images for this dataset were gathered by sampling the Kaggle **Dogs vs. Cats** images along with the **ImageNet** dataset for **panda** examples.

Dataset

Animals: Dogs, Cats, and Pandas



Figure: A sample of the 3-class animals dataset consisting of 1,000 images per dog, cat, and panda class respectively for a total of 3,000 images. Source: Adrain

Book

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- Just like MNIST, **CIFAR-10** is considered another standard benchmark dataset for image classification in the computer vision and machine learning literature.
- CIFAR-10 consists of 60,000 $32 \times 32 \times 3$ (RGB) images resulting in a feature vector dimensionality of 3072.
- As the name suggests, CIFAR-10 consists of 10 classes, including: airplanes, automobiles, birds, cats, deer, dogs, frogs, horses, ships, and trucks.
- Compared to $\geq 97\%$ classification accuracy on MNIST, it's substantially harder obtain such a model for CIFAR-10 and it's bigger brother, **CIFAR-100**

Dataset

CIFAR-10

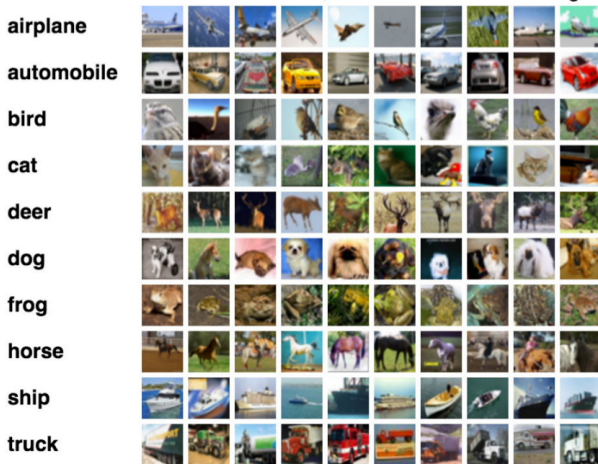


Figure: Example images from the ten class CIFAR-10 dataset

- The challenge comes from the dramatic variance in how **objects appear**. For example, we can no longer assume that an image containing a green pixel at a given (x, y) -coordinate is a frog. This pixel could be part of the background of a forest that contains a deer. Or, the pixel could simply be the color of a green truck.

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Dataset

SMILES

- Consists of images of **faces** that are either smiling or not smiling. In total, there are **13,165** grayscale images in the dataset, with each image having a size of 64×64 .
- Images in this dataset are tightly cropped around the face allowing us to devise machine learning algorithms that focus solely on the task of smile recognition.

Dataset

SMILES



Figure: Top: Examples of "smiling" faces. Bottom: Samples of "not smiling" faces. We will later train a Convolutional Neural Network to recognize between smiling and not smiling faces in real-time video streams. Source: Adrian Book.

- **Decoupling** computer vision preprocessing from machine learning (especially for benchmark datasets) is a common trend you'll see when reviewing popular benchmark datasets
- In some cases, it's unfair to assume that a machine learning researcher has enough exposure to computer vision to properly preprocess a dataset of images prior to applying their own machine learning algorithms.

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Dataset

Kaggle: Dogs vs. Cats

- The Dogs vs. Cats challenge is part of a Kaggle competition to devise a learning algorithm that can correctly classify an image as containing a **dog or a cat**.
- A total of **25,000** images are provided to train your algorithm with varying image resolutions.

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- The Flowers-17 dataset is a 17 category dataset with 80 images per class curated by Nilsback et al. [52]. The goal of this dataset is to correctly predict the species of flower for a given input image. A sample of the Flowers-17 dataset
- Flowers-17 can be considered a challenging dataset due to the dramatic changes in scale, view- point angles, background clutter, varying lighting conditions, and intra-class variation.
- As a general rule of thumb, it's advisable to have 1,000-5,000 example images per class when training a deep neural network

Dataset

Flowers-17

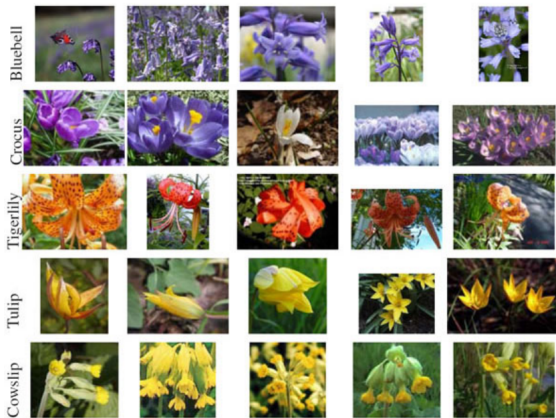


Figure: A sample of five (out of the seventeen total) classes in the Flowers-17 dataset where each class represents a specific flower species. Source: Adrian Book.

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Dataset

CALTECH-101

- the CALTECH-101 dataset is a popular benchmark dataset for object detection. Typically used for object detection (i.e., predicting the (x, y) -coordinates of the bounding box for a particular object in an image), we can use CALTECH-101 to study deep learning algorithms as well.
- The dataset of 8,677 images includes 101 categories spanning a diverse range of objects, including elephants, bicycles, soccer balls, and even human brains, just to name a few.

Dataset

Tiny ImageNet 200

- image classification challenge for students similar to the ImageNet challenge, but smaller in scope.
- There are a total of 200 image classes in this dataset with 500 images for training, 50 images for validation, and 50 images for testing per class. Each image has been preprocessed and cropped to $64 \times 64 \times 3$ pixels making it easier for students to focus on deep learning techniques rather than computer vision preprocessing functions.

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- The **Adience dataset**, Constructed in 2014, is used to facilitate the study of age and gender recognition. A total of 26,580 images are included in the dataset with ages ranging from 0-60. The goal of this dataset is to correctly predict both the age and gender of the subject in the image.

Dataset

Adience



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- ImageNet is actually a project aimed at labeling and categorizing images into almost **22,000** categories based on a defined set of words and phrases.
- there are over 14 million images in the ImageNet project.
- ImageNet follows the WordNet hierarchy [59]. Each meaningful word/phrase inside WordNet is called a “synonym set” or “synset” for short.
- Within the ImageNet project, images are organized according to these **synsets**, with the goal being to have **1,000+** images per synset

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- ImageNet is actually a project aimed at labeling and categorizing images into almost **22,000** categories based on a defined set of words and phrases.
 - 1 Angry
 - 2 Disgust (sometimes grouped in with “Fear” due to class imbalance)
 - 3 Fear
 - 4 Happy
 - 5 Sad
 - 6 Surprise

The End